



The Value of Lean

Kati Vilkki

Lean thinking brings significant advances to software development. So far, we've found useful in practice the focus on flow (instead of resource utilization), the role of managers, and tools for continuous improvement.

WHEN WE FIRST started experimenting with agile development at Nokia Networks in 2004, we talked about agile and iterative development. Since 2007, our approach has been “agile and lean” because lean thinking clearly brings some benefits in large-scale agile development, even though lean doesn't specify in any way the engineering practices or how to actually write software.

The point is lean thinking, not lean practices. Practices are always context specific; those that work well in manufacturing (which has repetitive tasks, low variability, and homogenous flow) often don't work at all in product development (which has nonrepetitive tasks, high built-in variability, nonhomogenous flow, and nonhomogenous cost of delay).

We learned some useful ideas from lean development:

- Focus on creating value for customers and optimize your whole system for it. This requires systems thinking and a holistic understanding of your system; all people involved must also understand it as well as its value to customers.
- Respect the people in the system: employees, customers, and vendors.
- Reduce cycle time to improve flow and to surface (and remove) waste. Trying to optimize resource allocation isn't the way to become efficient; instead, focus on managing your queues and limiting the work in progress, which will in turn reduce the cycle time.
- Build quality in. Integrate often, test early, and fix immediately.
- Create a continuous improvement culture. Everyone is responsible for doing their job and continuously thinking about how to do it better.

Reserve time for improvement in every iteration.

When all people in your organization have a common understanding of these key ideas, you gain a solid framework for improving the way the organization works.

In particular, understanding lean thinking helps managers improve the organization's management practices and structure. Lean also gives ideas about what a manager should do (coach, drive systematic improvement, and promote learning) and has helped many find a more meaningful role. Similarly, systems thinking and looking at the whole value stream improves cross-team cooperation. This is especially useful in product development, where software development is only one part. At the team level, lean thinking gives several useful tools for continuous improvement: understanding and removing waste, limiting work in progress, providing visual controls, speeding up feedback loops, systematically assisting in problem solving and improvement, and limiting work to capacity.

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Lean Is a Fad

Hakan Erdogmus

Lean thinking advances the maturity of software development, but it's by no means a silver bullet. Its real value is in concrete tools that development teams and managers can employ on the ground every day.

THERE, I SAID it. Even its most ardent advocates confirmed lean's fad status by using the tagline "The Next Wave of Software Process" at the 2009 Lean and Kanban Conference. Fads, like waves, come and go. But is lean just another fad? The short answer is a qualified "no." Let me explain.

A fad that is just another fad is useless, but not all fads are useless, and neither is lean. Lean is useful not as a revolutionary way of thinking but for suggesting concrete, alternative tools for software development. Without that aspect, lean is mostly new vocabulary: a set of mainstreamed concepts dressed in new clothes.

To see why lean isn't so novel, a quick comparison of its principles to the paradigm from which the lean movement is desperately trying to distinguish itself will suffice. The reference point I'm implying is the now-mainstream values

and principles of the Agile Manifesto. To represent lean, we can use Poppendieck's seven-element version augmented with James P. Womack and Daniel T. Jones's people-process-purpose (3Ps) triad (*Lean Thinking*, 2nd ed., Simon & Schuster, 2003). Within both agile and lean approaches, variations of tenets exist, but the differences are due to whether something is deemed as a tenet or a corollary of one. If we avoid nitpicking based on semantics, it's easy to come up with a many-to-many mapping such that the transitive closures of the two approaches nearly converge. Although some principles might be elemental in one approach, others can be inferred. Ultimately, we don't end up at fundamentally different places regardless of the starting point. Only a few residual, apparent differences remain, pertaining primarily to flexibility, systems thinking, and process focus.

At first glance, the emphasis granted to flexibility in agile is missing in lean, but a quick review of the lean literature reveals much discussion on flexibility and how lean principles support it. Why? Because flexibility is a desired property that can naturally be inferred from lean's main concepts, learning and continuous improvement, in particular.

What's apparently unique in lean is systems thinking, the predominance of the whole. Systems thinking is characterized by a long-term view and a balanced focus that together favor the sum over the parts. But is it as unique and as useful as it appears? First, systems thinking is mostly ignored in actual software practice as a governing principle, despite much lip service. Second, it's implicit in existing paradigms, particularly in agile, although more at the team level than at an organizational level. In real practice, the difference isn't noteworthy.

Lean thinking has a strong process focus, something that is deceptively missing in agile. I prefer an implicit process focus to an explicit one. The latter can be highly counterproductive in low-fidelity implementations by giving an overriding

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So is lean just another management fad? Is Kanban the next hype after Scrum? Yes and no. Yes, because I'm sure we'll see more Kanban in the upcoming years—that is, people applying a process or making a process change without understanding the thinking behind it and without making any fundamental changes. And no, because some organizations and teams will manage to become very lean, change the underlying mental model, and continuously improve.



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Vilkki and I agree on two points: lean is a worthy trend, and lean could be a fad based on that perspective. But we disagree on why it could be a fad—or more fundamentally, where the real value in lean lies. I maintain that it's in the concrete practices and tools. Vilkki claims that it's in the underlying philosophy, the lean thinking. In past management fads, I've seen the big-picture thinking come and go, shift subtly, or change color depending on the political priorities of the day. Practices and tools persist much longer. They get entrenched in the grassroots, habitualized even when the so-called mental thinking turns over, with ivory tower visionaries latching onto something else that sounds better. Recycling is the perfect example of a persistent practice. In our industry, practices and tools have an even higher chance of surviving hype waves than in other industries because the people close to the ground are both empowered and

intellectually capable, hence they're well positioned to operationalize the underlying thinking and fine-tune the philosophy to fit the context. Besides, the big-picture thinking often simply echoes the already universal values. Consider the key lean thinking ideas Vilkki (and I) find useful: focus on value creation, respect people, reduce cycle time, build quality in, and continuously improve. Who could object to any of them? But did we not know about the role of each of these ideas before lean thinking came along? We certainly did, for countless management trends, including agile, have stressed them and advocated them explicitly using different vocabulary. Each idea embodies a classical textbook management principle. If they didn't stick before, they won't just because they are now collectively called lean thinking. The real meat is ultimately in the tools and practices that address the difficult trade-offs when the philosophy hits the ground.

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importance to process. A misguided effect can easily result when systems thinking is forgotten and isn't able to dampen that importance in favor of balance and optimizing the whole. Philosophical guidance isn't enough to enforce the tenets. We need solid, practical tools suited to the software context to help.

That's why when evaluating a new process approach, I tend to focus on concrete practices. Philosophy and values are fine, but we're demonstrably inconsistent in maintaining and evaluating such soft targets. Practices are concrete manifestations of the fundamentals—and more measurable. In that sense, lean does offer some new tools to the software community, of

which Kanban and value-stream mapping are main examples. The former, Kanban, provides an alternative and a complement to iteration-based workflow management. The latter, VSM, provides analytic techniques for identifying and managing pain points. VSM isn't yet lovingly embraced in the software industry: it just doesn't appear to be the low-hanging fruit that Kanban is. But Kanban is spreading fast and filling a gap left by leading agile implementations. After the next wave supplants lean, Kanban will probably stick around. Lean is the new agile, but Kanban is not the new Scrum. Tools are cumulative while philosophies are replaceable.

In less than a decade, a next wave of software process will hit, and it will be the new lean. Will that wave be a fad, too? By definition, yes, but hopefully it won't be just another fad. ☞



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VILKKI RESPONDS

There seems to be a pattern in how new ideas and practices come to be: first, there are early experiments, with most people saying "that will never work." Then comes gradually growing enthusiasm and (management) hype: "This will solve all our problems." Finally, we get to realism about which ideas and practices are truly helpful and stick and how difficult it is to make significant improvement. Of course, many ideas and practices never make it this far and die on the way (we might call these "just another fad").

I think agile has passed the chasm to realism at least to some extent, and many agile practices are now well proven in practice (for example, continuous integration and test early). It's a long time since I last heard someone propose, "Let's integrate and verify the product only once at the end."

The most influential practices are those that are directly connected to doing the real work—to creating value for the customer, not management practices. Time will show whether Scrum will be around in 10 years' time (personally, I doubt it), but I'm willing to bet that iterative development, continuous integration, test early, and self-organized teams are here to stay.

Whether lean will be just another fad or make a real difference depends on how well its concepts and practices improve our ability to create value for the customer. Based on our experience at Nokia, I claim that limiting works in progress, improving visual controls, systematically tackling problem-solving, and looking at the whole value stream are truly powerful practices that are very difficult to implement. Thus lean has the potential to be more than just another fad.